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Case Report

Preemptive, Self-Applied Photobiomodulation in the Treatment of Oral Mucositis: A High-Risk Case Study

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Keywords
Oral mucositis · Stomatitis · Photobiomodulation · Head and neck cancer · Case report

Abstract
Introduction: Oral mucositis (OM) is a prevalent and debilitating complication of antineoplastic therapy and hematopoietic stem cell transplantation. Effective OM treatment has important economic, quality of life (QoL), and healthcare implications. Photobiomodulation (PBM) emerges as an innovative addition to the treatment paradigm, using red and near-infrared light to relieve pain and stimulate tissue repair without significant toxicity. Although the role of PBM is well established in international guidelines, optimal timing and delivery of therapy remain unclear. Several studies have reported encouraging data, advocating for a preemptive and self-applied approach to PBM. ***Case Presentation:*** We describe a high-risk case of locally advanced (T2N1M0, stage II) nasopharyngeal squamous cell carcinoma. The patient was treated with primary chemoradiation consisting of weekly cisplatin and a radiation dose of 70 Gy in 35 fractions to the nasopharynx, lymph nodes, and surrounding soft tissues. As per the 904 nm laser lymphatic protocol, 3J/point was delivered over 30 s at an output power of 200 mW in the continuous mode, targeting the tongue, gingiva, buccal mucosa, and hard palate. Various clinical outcome measures were documented including self-reported health status (PROMS), oral health-related quality of life (OHIP-14), and objective OM severity (OMAS). Owing to raised PROMS and OHIP-14 scores on Mondays reflecting symptom exacerbation over the weekend, the study protocol was adjusted at day 21 to include a portable SYMBYX 600 nm laser for the patient to self-apply over the weekend. In contrast to expected outcomes for concurrent chemoradiation, the patient developed only mild to moderate OM: PROMS (28/100), OHIP-14 (19/56), and OMAS (6/45). Further, the patient did not require regular analgesia, opioid

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analgesics, or gastrostomy tube placement. ***Conclusion:*** Preemptive PBM using the 904 nm laser lymphatic protocol, supplemented by a self-applied SYMBYX 600 nm laser to facilitate continued treatment over the weekends, shows promise as an effective therapy for reducing OM severity in high-risk patients. Further studies are required to clarify optimal timing and delivery of PBM within the OM treatment paradigm.

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Introduction

Oral mucositis (OM) can be understood as erosive inflammation of the oropharyngeal mucosa, resulting in pain, dysphagia, and decreased oral intake [1]. OM is a leading complication of antineoplastic therapy and hematopoietic stem cell transplantation (HSCT), with important economic, psychosocial, and healthcare implications [2, 3]. The pathogenesis of OM begins with the formation of highly reactive oxygen species within the basal cell layer of the mucosa, resulting in upregulation of proinflammatory cytokines and secondary tissue damage [4]. OM is common and is seen in 20–40% of patients undergoing antineoplastic treatment for solid tumors and 80–99% of patients undergoing HSCT [2, 5]. The incidence of OM has been reported to be as high as 100% in patients receiving high-dose radiotherapy (RT; >50 Gy), with around 85% developing World Health Organization (WHO) grade III or IV OM [6–8].

Effective OM treatment has important economic, psychosocial, and healthcare implications. OM has been linked to increased hospital stay, exerting an economic burden on healthcare systems [9]. Additionally, OM is known to cause considerable oral pain, which is the main factor affecting patients' quality of life (QoL) [10]. Research has demonstrated a sharp decline in QoL of patients with head and neck cancer following the initiation of therapy, emphasizing the need for rigorous symptomatic control [11]. Furthermore, the high symptom burden associated with OM is often responsible for treatment adjustments or cessation, compromising treatment efficacy and potentially increasing mortality rates [2, 3]. The standard of care for OM has classically centered on supportive care, including opioid analgesia, which places the patient at risk of various toxicities including sedation, nausea, vomiting, and delirium [11]. Photobiomodulation (PBM) emerges as an innovative addition to OM treatment, using red and near-infrared light to relieve pain and stimulate tissue repair without significant toxicity, especially when administered preemptively [12]. Although the role of clinic-based PBM is well established in international guidelines, optimal timing and delivery of therapy remain unclear [13–15]. Notably, a self-applied PBM protocol has recently received empirical support [16, 17]. This novel approach aims to optimize therapeutic benefits of clinic-based PBM through increased accessibility. A case of locally advanced nasopharyngeal carcinoma is described, in which concurrent chemotherapy and RT exceeding 50 Gy posed a high risk of OM. Preemptive, self-applied PBM, using a portable device to facilitate weekend use, has been shown to be effective in reducing OM severity compared with anticipated outcomes from the literature.

Case Presentation

A 52-year-old male was diagnosed with locally advanced (T2N1M0, stage II) nasopharyngeal squamous cell carcinoma. His treatment regimen included a 7-week course of concurrent cisplatin (50 mg/mL/week) and RT, with doses of 70 Gy in 35 fractions to the right

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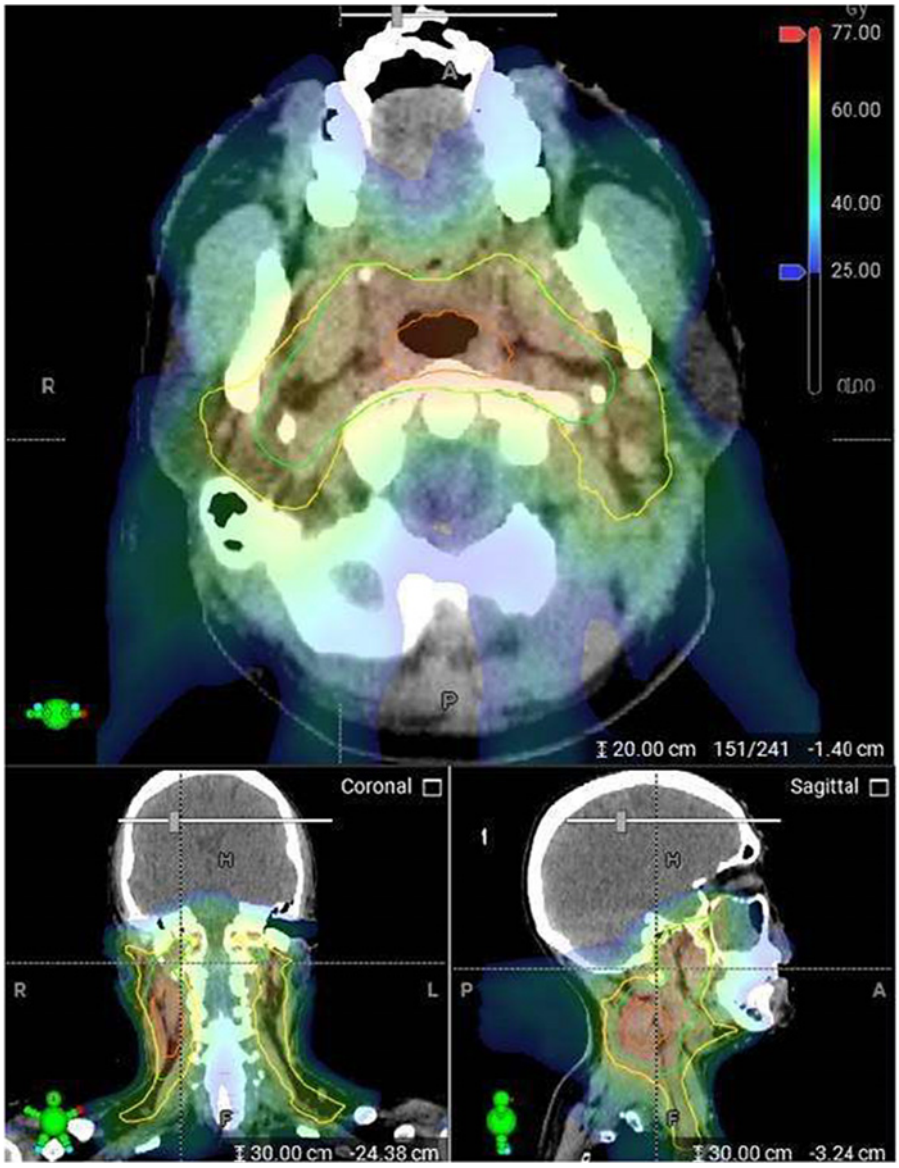


Fig. 1. Planning CT with overlying dose wash, showing the case study's nasal cavity, oral cavity, and bilateral cervical lymph nodes were exposed to a radiation dose as high as 77 Gy.

nasopharynx, 63 Gy in 35 fractions to FDG-PET avid lymph nodes and surrounding soft tissue, and 56 Gy in 35 fractions to the left parotid and supraclavicular fossa. Figure 1 illustrates the planning CT with dosewash as a reference. Prior to commencing antineoplastic treatment, the clinician administered PBM after obtaining informed consent from the patient. As per the 904 nm laser lymphatic protocol, 3J/point was delivered over 30 s at an output power of 200 mW in the continuous mode, targeting the tongue, gingiva, buccal mucosa, and hard palate. The clinician administered PBM within an hour of RT. Throughout treatment, the patient reported their self-perceived health status (patient-reported outcome measure [PROMS]) and QoL (oral health quality of life [OHIP-14]) [17, 18]. Additionally, the clinician assessed the Oral Mucositis Assessment Score (OMAS) to gauge OM severity [19]. The CARE Checklist has been completed by the authors for this case report, attached as online supplementary material (for all online suppl. material, see https://doi.org/10.1159/000541966).

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Fig. 2. A series of clinical photographs of the oral cavity throughout RT. Top left: left lateral tongue ulceration with generalized erythema at session 9. Top right: left lateral tongue ulceration resolving at session 43. Middle left: slightly erythematous right buccal mucosa at session 18. Middle right: normal right buccal mucosa at session 46. Bottom left: erythematous soft and hard palate with ulceration and pseudomembrane formation at session 20. Bottom right: soft and hard palate displaying improved erythema with resolution of ulceration at session 46.

The patient received 35 fractions of RT from Monday to Friday for 7 weeks. Following the completion of RT on day 46, the patient's clinical outcome measures were monitored for an additional 3 weeks. Clinically, the initial signs of OM had surfaced during the ninth session. At this time, the patient exhibited mild erythema on the soft and hard palates, along with a 2 × 3 mm ulcer on the left lateral tongue, accompanied by intermittent odynophagia. These changes are shown in Figure 2 (top left). By the 18th session, mild erythema was visible on the buccal mucosa, and odynophagia had become more persistent, as illustrated in Figure 2 (middle left). By session 20, notable erythema and ulceration appeared on the soft and hard palate, as shown in Figure 2, bottom left. Despite this, only minor dietary modifications are required, including avoidance of extreme temperatures and spicy foods.

Days 30–40 marked the peak of the patient's discomfort and functional limitations, as shown in Figure 3. For each clinical outcome measure, higher scores reflect greater severity.

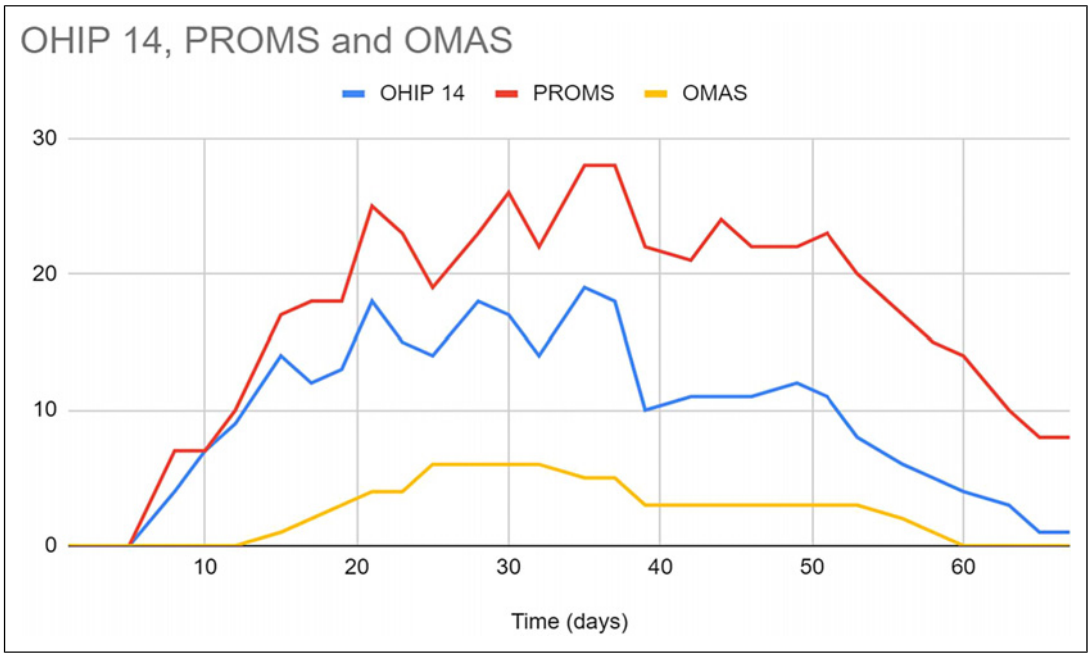


Fig. 3. Clinical outcome measures (OHIP-14, PROMS, and OMAS) over time. The case study received 35 fractions of RT Monday to Friday for 7 weeks. Allowing for weekends, RT was completed on day 46. The clinical outcome measures were recorded for a further 3 weeks following the cessation of RT, finishing on day 67.

Overall, the clinical impact was modest: PROMS (28/100), OHIP-14 (19/56), and OMAS (6/45). PROMS and OHIP-14 scores were higher on Mondays, reflecting symptom exacerbation over weekends when clinic-based PBM was unavailable. The study protocol was subsequently adjusted on day 21 to include a portable SYMBYX 600 nm laser for the patient to self-apply over the weekend. The clinician provided education regarding its use and provision was given to allow direct application to the soft palate. Gradual clinical improvement, if not complete resolution, was evident in all oral subsites (soft and hard palate, left lateral tongue, and buccal mucosa) by session 46, as depicted in Figure 2 (top, middle, and bottom right). All outcome measures markedly improved by week 3 post-RT (8/100 PROMS, 1/56 OHIP-14, and 0/45 OMAS). Overall, PBM was well tolerated with no reported adverse effects. At 12-month follow-up, there was no evidence of late toxicity including radiation fibrosis.

Discussion

High-dose RT (>50 Gy) and concurrent chemotherapy placed our case at high risk of developing OM. Existing OM treatment paradigms are often clinically inadequate, prioritizing reactive over proactive treatment. Further, clinic-based interventions compromise accessibility of treatment and overall therapeutic benefit. Recent innovations in PBM advocating for a preemptive and self-applied approach have received empirical support [16, 17]. In our study, we assessed the role of PBM, incorporating a portable device to facilitate continuous treatment over the weekend, in reducing OM severity. This marks the first Australian study evaluating self-applied PBM in a patient with head and neck cancer. Overall, our patient displayed greatly reduced OM severity compared with expected outcomes. Furthermore, PBM was well tolerated with no reported adverse effects.

RT-induced OM is known to occur in the majority of head and neck cancer patients, with close to 100% of patients treated with high-dose RT (>50 Gy) [8]. Elting et al. [2] reported that 99% of patients with oropharyngeal tumors treated with high-dose radiation had OM, and 85% of these patients had severe OM. In our patient treated with PBM, however, the peak clinical outcomes for self-perceived health status, QoL, and objective OM severity were moderate at most (PROMS 28/100, OHIP-14 19/56, OMAS 6/45). Notwithstanding these encouraging results, PBM did not appear to delay the onset of OM. A prospective observational study involving 30 patients with head and neck cancer receiving high-dose RT revealed that 43% experienced OM within the first week, escalating to 100% by the third week [7]. Despite undergoing PBM therapy, our patient developed clinically significant OM within the initial 3 weeks of radiation treatment. These collective findings suggest that PBM may assist in mitigating the severity of OM but does not seem to postpone the onset of symptoms.

The requirement for analgesia and enteral feeding also informs treatment success. According to the existing literature, pain typically escalates by the third week, reaches its peak by the fifth week, and typically subsides within 2–6 weeks post-therapy [20]. This pain is often severe and impairs important biological functions such as swallowing. A study examining patients treated with high-dose radiation therapy for oropharyngeal tumors found that 90% of patients required regular analgesics, with 65% relying on opioids [2]. Our patient did not require opioid analgesics, mitigating complications linked to their adverse effects. Additionally, 58% required gastrostomy tube (G-tube) insertion due to OM-related complications [2]. In our case, prophylactic G-tube placement was averted owing to promising early treatment outcomes. The patient maintained oral feeding throughout RT, necessitating only minor dietary adjustments.

The impact of OM on QoL is profound, with oral pain being the driving factor [7, 11]. Consequently, the QoL of patients with head and neck cancer has been shown to decline markedly after receiving antineoplastic treatment [11]. Although our patient demonstrated a gradual increase in PROMS throughout the study, these scores remained relatively low and declined quickly following completion of RT. We also observed exacerbation of PROMS every Monday, reflecting the lack of PBM delivery over the weekend. However, after the transition to self-applied PBM on day 21, these Monday PROMS exacerbations decreased, as depicted in Figure 3. In line with previous studies investigating self-applied PBM, this corroborates the use of a portable device in providing pain relief and accelerating tissue healing [16, 17].

Conclusion

In summary, OM remains a prevalent and debilitating complication of antineoplastic therapy and HSCT. We believe that an important shift in the OM treatment paradigm is required. Preemptive PBM, transitioning from clinic-based to self-applied, is required to further optimize therapeutic benefits. Within the inherent limitations of a case study, we posit that preemptive, self-applied PBM shows promise as an effective therapy for reducing OM severity in patients undergoing high-dose RT. Notably, PBM was effective in enhancing all outcome measures, including self-perceived health status, QoL, and objective OM severity. We believe this case study will direct further research aimed at clarifying the optimal timing and delivery of PBM within the OM treatment paradigm.

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Statement of Ethics

Ethics approval was obtained from The Royal Children's Hospital Melbourne Ethics Committee in 2019 (confirmation ID: HREC 2019.104). Additional ethics approval was obtained through the Sydney Adventist Hospital as a feasibility quality improvement progressive through the office of the Medical & Clinical Governance. Written informed consent was obtained from the patient for the publication of this case report and any accompanying images.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

J.R. was responsible for the concept of the case report, obtaining patient consent, compiling medical images, and performing data analysis and interpretation. L.N.C. provided the input for the design, data analysis, and interpretation. R.J.D. assisted with data analysis and interpretation. All authors contributed to the drafting of the manuscript.

Data Availability Statement

All the data analyzed in this case study will be included in this article and online supplementary files. Further inquiries can be directed to the corresponding authors.

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